Experiment 7

Q1 Answer following question in typed form and paste it in the text box provided for submission: **bold text**

**1 What is need for Hypothesis Testing ? **

Need for Hypothesis Testing

Hypothesis testing is a statistical method used to make decisions or inferences about a population based on sample data.

Why do we need Hypothesis Testing? ** 1. Decision Making** Helps to decide whether a claim or assumption is true or false Example: Checking if a new medicine is effective

2. Drawing Conclusions from Sample Data .We usually cannot study the whole population So, we take a sample and use hypothesis testing to generalize results

3. Testing Assumptions (Claims) Used to test statements like: Mean = some value Two groups are equal Example: Comparing student marks of two classes

4. Reducing Uncertainty .Provides a scientific and systematic method instead of guessing .Uses probability to support decisions

5. Comparing Groups .Helps to check whether difference between groups is significant or due to .chance .Example: .Before vs After experiment .Control vs Treatment group

6. Basis for Statistical Analysis & Research .Widely used in: .Data Science .Machine Learning .Medical research .Business decisions

2 With a neat diagram show important steps in hypothesis Testing. **bold text**

Important Steps in Hypothesis Testing (with Diagram)

Neat Diagram (Flow)

Start ↓ Define Problem ↓ Formulate Hypotheses (H0: Null, H1: Alternative) ↓ Select Significance Level (α) ↓ Choose Appropriate Test (Z-test, t-test, etc.) ↓ Collect Data & Compute Test Statistic ↓ Find p-value / Critical Value ↓ Make Decision (Reject or Fail to Reject H0) ↓ Conclusion

Explanation of Steps

1. Define the Problem .Clearly state what you want to test .Example: Check if average marks are greater than 50

2. Formulate Hypotheses Null Hypothesis (H_0): No effect / no difference Alternative Hypothesis (H_1): There is an effect or difference

👉 Example:

H_0: $\mu = 50$ H_1: $\mu \neq 50$

3. Select Significance Level (α) Common values: 0.05 (5%) or 0.01 (1%) It defines the risk of rejecting true H_0

4. Choose Statistical Test Based on data type: Z-test t-test Chi-square test ANOVA

5. Collect Data & Compute Test Statistic Calculate value like: Z-value t-value

6. Find p-value / Critical Value Compare calculated value with: Table value OR p-value

7. Make Decision If p-value ≤ α → Reject H_0 If p-value > α → Fail to Reject H_0

8. Conclusion State final result in simple words Example: "There is significant difference"

3 What is a student t distribution?

Scanned with OKEN Scanner

Student's t-Distribution

Student's t-distribution is a type of probability distribution used in statistics when:

The sample size is small (usually n < 30 n<30) The population standard deviation is unknown

It was introduced by William Sealy Gosset, who used the name "Student," hence the name t-distribution.

Definition

The Student's t-distribution is a symmetric, bell-shaped distribution similar to the normal distribution but with heavier tails.

Key Features

1. Bell-Shaped Curve Looks like normal distribution Symmetrical around the mean ** 2. Heavier Tails** More spread out than normal distribution Higher probability of extreme values

3. Depends on Degrees of Freedom (df) $d\ f$

n − 1 df=n−1 Shape changes with sample size: Small sample → wider curve Large sample → approaches normal distribution

4. Used When: Sample size is small Population variance is unknown

Applications One-sample t-test Two-sample t-test Confidence intervals for mean

Example

If you take a sample of 10 students and want to estimate the average marks of all students, you will use the t-distribution instead of normal distribution.

4 What is level of significance ? What role t plays in hypothesis Testing

Level of Significance (α)

Definition

The level of significance (α) is the probability of rejecting the null hypothesis (H_0) when it is actually true.

Common Values α

0.05 α=0.05 → 5% risk α

0.01 α=0.01 → 1% risk

Importance It sets the decision boundary in hypothesis testing Helps decide whether a result is statistically significant or not

** Rule:**

If p-value ≤ α → Reject H_0 If p-value > α → Fail to reject H_0

Role of t in Hypothesis Testing ** What is t-value?**

The t-value is a test statistic calculated from sample data to test hypotheses about the population mean.

Formula

t

$x\bar{}-\mu\ s/n$ t= s/ n

x ¯ −μ

Role of t

Scanned with OKEN Scanner

1. Measure Difference Shows how far the sample mean is from the population mean Measured in terms of standard error ** 2. Decision Making** Compare calculated t-value with: Critical t-value (from table), OR p-value

** Decision:**

Large t-value → strong evidence against H_0 Small t-value → weak evidence **3. Used for Small Samples** Applied when: Sample size is small Population standard deviation is unknown **4. Helps in t-tests** One-sample t-test Two-sample t-test Paired t-test

5 Design a hypothesis testing for identifying average age of students in TE IT class is 19 or not. **bold text**

Hypothesis Testing Design (Average Age of TE IT Students)

Problem Statement

To test whether the average age of TE IT students is 19 years or not.

Step 1: Formulate Hypotheses Null Hypothesis (H_0):

μ

19 μ=19 (Average age is 19) Alternative Hypothesis (H_1):

$\mu \neq 19$ μ □ =19 (Average age is not 19)

This is a two-tailed test

** Step 2: Select Level of Significance** Choose: α

0.05 α=0.05 (5%) ** Step 3: Choose Test** Use t-test because: Sample size is small Population standard deviation is unknown ** Step 4: Test Statistic**

t

$\bar{x} - 19$ s / n t= s/ n

$\bar{x} - 19$

Where:

$\bar{x}$ x ¯ = sample mean age s s = sample standard deviation n n = sample size ** Step 5: Decision Rule** Find critical t-value from table using: $d\ f$

$n - 1$ df=n−1 α

0.05 α=0.05

Decision:

If $|\ t\ |$

$t\ c\ r\ i\ t\ i\ c\ a\ l$ |t|>t critical

→ Reject H_0 If $|\ t\ | \leq t\ c\ r\ i\ t\ i\ c\ a\ l$ |t|≤t critical

→ Fail to reject H_0 **Step 6: Conclusion** If H_0 is rejected: **Average age is not 19** If H_0 is not rejected: **No significant difference (average age ≈ 19)**

6 Check if you can accept Ho at alpha 95% ? Match your conclusions with python functions. **bold text**

**To check whether we accept ** H 0 H 0

at 95% confidence (α = 0.05) and verify using Python, we need a sample dataset (ages of students).

Since no data is given, I'll show a complete method with example + Python verification (you can replace values with your actual data).

Step 1: Given (Example Data)

Assume:

Scanned with OKEN Scanner

Sample size: n

10 n=10 Sample mean: $\bar{x}$

19.2 x ¯ =19.2 Standard deviation: s

1.0 s=1.0 Hypothesized mean: μ

19 μ=19 α

0.05 α=0.05 (95% confidence)

Step 2: Hypotheses H 0 : μ

19 H 0

:μ=19 H 1 : $\mu \neq$ 19 H 1

:μ □ =19 (two-tailed test)

Step 3: Calculate t-value

t

19.2 − 19 1 / 10 t= 1/ 10

19.2−19

t

0.2 0.316 ≈ 0.63 t= 0.316 0.2

≈0.63

Step 4: Critical Value Degrees of freedom: $d\ f$

9 df=9 At α

0.05 α=0.05 (two-tailed):

$t\ c\ r\ i\ t\ i\ c\ a\ l$ ≈ ± 2.262 t critical

≈±2.262

Step 5: Decision Calculated t

0.63 t=0.63 0.63 < 2.262 0.63<2.262

** Fail to reject ** H 0 H 0

Step 6: Conclusion

At 95% confidence level, we accept H 0 H 0

There is no significant difference, so average age ≈ 19

```
import numpy as np
from scipy import stats

# Sample data (example)
data = [18, 19, 20, 21, 19, 18, 20, 19, 19, 20]

# Perform one-sample t-test
t_stat, p_value = stats.ttest_1samp(data, 19)

print("t value:", t_stat)
```


Scanned with OKEN Scanner

```
print("t-value:", t_stat)
print("p-value:", p_value)

# Decision
alpha = 0.05
if p_value <= alpha:
    print("Reject H0")
else:
    print("Fail to reject H0")
```

```
t-value: 1.0000000000000024
p-value: 0.34343639613791216
Fail to reject H0
```

Close

```
**Python Result Interpretatio**n
If p-value > 0.05 → Accept
H
0
H
0


If p-value ≤ 0.05 → Reject
H
0
H
0



** In this case:**

p-value will be greater than 0.05
So H₀ is accepted
```

Python Result Interpretation If p-value > 0.05 → Accept H 0 H 0

If p-value ≤ 0.05 → Reject H 0 H 0

** In this case:**

p-value will be greater than 0.05 So H₀ is accepted

Scanned with OKEN Scanner